**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 6 · STANDARD 6.AT.7**

# Simplify Algebraic Expressions

Lesson 6-15 · Teacher Copy

**Name:** Type your name **Date:** Today's date **Class:** Class period

## My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can simplify algebraic expressions by combining like terms.**

**Language Objective: I can explain my steps using the words like terms, combine, simplify, and coefficient.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Like Terms | Combine | Simplify | Coefficient | Term | Distributive property

☝️ *Tap any word to see its meaning and a picture.*

**1** Terms with the same variable part, like 3y and 8y, are _____.

**2** To add or subtract like terms into a single term is to _____ them.

**3** Rewriting an expression in a shorter form that is still equal to it is to _____ it.

**4** The number multiplied by a variable is the _____.

**5** A number or variable separated by + or − signs is a _____.

**6** Multiplying each part of a sum by a number uses the _____ property.

## Write About the Math

**Simplify the total order expression. What does each part of the simplified expression tell you?**
***Simplifica la expresión del pedido total. ¿Qué te dice cada parte de la expresión simplificada?***

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Like Terms** — *Términos semejantes*
- ☐ **Combine** — *Combinar*
- ☐ **Simplify** — *Simplificar*
- ☐ **Coefficient** — *Coeficiente*
- ☐ **Term** — *Término*

**Model:** *You can only combine items of the same type, just like like terms.* — Students explain microphones combine with microphones (4 + 2 + 6 = 12) but not with stands, connecting this to combining like terms in algebra.

### Start

Complete the frames. Use the word bank.
*Completa las oraciones. Usa el banco de palabras.*

- **The total order simplifies to ____m + ____. The ____ tells us ____ and the ____ tells us ____.**
- **My answer is ____ because ____.**
  *Mi respuesta es ____ porque ____.*

### Build

Write two connected sentences. Use because, so, or but.
*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **My evidence is ____, so ____.**
  *Mi evidencia es ____, entonces ____.*

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

## Answer Key & Teacher Guide

1. **Notes 1:** Like Terms
2. **Notes 2:** Combine
3. **Notes 3:** Simplify
4. **Notes 4:** Coefficient
5. **Notes 5:** Term
6. **Notes 6:** Distributive property
7. **Watch for this mistake:** *A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms — for example, simplifying 6x + 4 + 2x as 12x by adding all three numbers (6 + 4 + 2 = 12) instead of recognizing that 4 has no variable and cannot combine with 6x and 2x. The correct simplified expression is 8x + 4 (combine 6x + 2x = 8x, and let the 4 stay on its own). Before you submit, check that every term you combined has the exact same variable part.*
8. **Try It 1:** A. 4y — *Subtract the coefficients of the like terms: 7 − 3 = 4. Keep the variable y, giving 4y.*
9. **Try It 2:** A. 9m + 4 — *6m + 3m = 9m. The constant 4 has no like term, so it stays. Result: 9m + 4.*